



FreeRTOS in STM32CubeIDE

PG-DESD @ SUNBEAM INFOTECH

New Alt+Shift+N >

Open File...

Open Projects from File System...

Recent Files >

Close Editor Ctrl+W

Close All Editors Ctrl+Shift+W

Save Ctrl+S

Save As...

Save All Ctrl+Shift+S

Revert

Move...

Rename... F2

Refresh F5

Convert Line Delimiters To >

Print... Ctrl+P

Import...

Export...

Properties Alt+Enter

Switch Workspace >

Restart

Exit

Makefile Project with Existing Code

C/C++ Project

STM32 Project

STM32 Project from an Existing STM32CubeMX Configuration File (.ioc)

Project... Create a New STM32 project

Source Folder

Folder

Source File

Header File

File from Template

Class

Other... Ctrl+N

0 items

Description	Resource	Path	Location

Memory Regions		Memory Details			
Region	Start address	End address	Size	Free	Used

Target Selection

Select STM32 target or STM32Cube example

- MCU/MPU Selector
- Board Selector
- Example Selector
- Cross Selector

Board Filters

★

Commercial Part Number

STM32F407G-DISC1

Vendor

>

Type

>

Check/Uncheck All

☒ Discovery Kit

☐ Evaluation Board

☐ Evaluation Kit

☐ Nucleo USB Dongle

☐ Nucleo-144

☐ Nucleo-32

☐ Nucleo-64

☐ Nucleo-RF Kit

Features

Large Picture

Docs & Resources

Datasheet

Buy

STM32F4 Series

★

STM32F407G-DISC1

ACTIVE

Active

Product is in mass production

Part Number : STM32F4DISCOVERY

Commercial Part Number : STM32F407G-DISC1

Unit Price (US\$) : 19.89

Mounted Device : [STM32F407VGTx](#)

The STM32F4DISCOVERY Discovery kit leverages the capabilities of the STM32F407 high-performance microcontrollers, to allow users to develop audio applications easily. It includes an ST-LINK/V2-A embedded debug tool, one ST-MEMS digital accelerometer, one digital microphone, one audio DAC with integrated class D speaker driver, LEDs, push buttons, and a

Boards List: 1 item

Export

STM32 Project

Setup STM32 project

Project

Project Name:

☒ Use default location

Location:

Options

Targeted Language

☒ C ☐ C++

Targeted Binary Type

☒ Executable ☐ Static Library

Targeted Project Type

☒ STM32Cube ☐ Empty

IDE

Build Analyzer Static Stack Analyzer

Memory Regions Memory Details

Region	Start address	End address	Size	Free	Us

Problems Tasks Console Properties

0 items

Description	Resource	Path	Loca

STM32 Project

Firmware Library Package Setup

Setup STM32 target's firmware

Target and Firmware Package

Target Reference: STM32F407G-DISC1

Firmware Package Name and Version: STM32Cube FW_F4 V1.26.2

Firmware and Software Package Repository

Location: C:\Users\sunbeam\STM32Cube\Repository

See ['Firmware Updater'](#) for settings related to package installation

Code Generator Options

☐ Add necessary library files as reference in the toolchain project configuration file

☐ Copy all used libraries into the project folder

☒ Copy only the necessary library files

?

< Back

Next >

Finish

Cancel

Problems 0 items

Description	Resource	Path	Location
-------------	----------	------	----------

Build Analyzer Static Stack Analyzer

Memory Regions

Region	Start address	End address	Size	Free	Used
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STM32 Project

Firmware Library Package Setup

Setup STM32 target's firmware

Target and Firmware Package

Target Reference: STM32F407G-DISC1

Firmware Package Name and Version: STM32Cube FW_F4 V1.26.2

Board Project Options:

?

Initialize all peripherals with their default Mode ?

YesNo

Code Generator Options

☐ Add necessary library files as reference in the toolchain project configuration file

☐ Copy all used libraries into the project folder

☒ Copy only the necessary library files

Perform Project Creation. Please Wait For Completion ...

?

< BackNext >FinishCancel

ProblemsTasksConsoleProperties

0 items

Description	Resource	Path	Location

Build AnalyzerStatic Stack Analyzer

Memory RegionsMemory Details

Region	Start address	End address	Size	Free	Used

STM32 Project

Firmware Library Package Setup

Setup STM32 target's firmware

Target and Firmware Package

Target Reference: STM32F407G-DISC1

Firmware Package Name and Version: STM32Cube_FW_F4 V1.26.2

Open Associated Perspective?

Device Configuration Tool editor is associated with Device Configuration Tool perspective. Do you want to open this perspective now?

☒ Remember my decision

Yes No

Code Generator Options

☐ Add necessary library files as reference in the toolchain project configuration file

☐ Copy all used libraries into the project folder

☒ Copy only the necessary library files

Perform Project Creation. Please Wait For Completion ...

< Back Next > Finish Cancel

Problems Tasks Console Properties

0 items

Description	Resource	Path	Location

Build Analyzer Static Stack Analyzer

Memory Regions Memory Details

Region	Start address	End address	Size	Free	Used

Pinout & Configuration

Clock Configuration

Project Manager

Tools

Software Packs

Pinout

- Undo Mode and pinout Ctrl-Z
- Redo Mode and pinout Ctrl-Y
- ☐ Keep Current Signals Placement Ctrl-K
- ☒ Show User Label
- Disable All Modes Ctrl-D
- Clear Pinouts Ctrl-P
- Clear Single Managed Signal Ctrl-M
- Pins/Signals Options... Ctrl-O
- List Pinout Compatible MCUs Alt-L
- Export pinout with Alt. Functions
- Export pinout without Alt. Functions Ctrl-U
- Reset used GPIOs Alt-G
- Set unused GPIOs Ctrl-G
- Pinout View Colors
- Layout reset

LQFP100

- IDE leds
 - > Includes
 - > Core
 - > Inc
 - > Src
 - > Startup
 - > Drivers
 - > CMSIS
 - > Device
 - > Include
 - > STM32F4xx_HAL_Driver
 - MX leds.ioc
 - STM32F407VGTx
 - STM32F407VGTx

Categories A-Z

- System Core
- Analog
- Timers
- Connectivity
- Multimedia
- Security
- Computing
- Middleware

FATES

- ☒ FREERTOS
- LIBJPEG
- LWIP
- MBEDTLS
- PDM2PCM
- USB_DEVICE
- USB_HOST

FREERTOS Mode and Configuration

Mode

Interface CMSIS_V1

Configuration

Reset Configuration

- ☒ Mutexes
- ☒ Events
- ☒ FreeRTOS Heap Usage
- ☒ Tasks and Queues
- ☒ Timers and Semaphores
- ☒ Advanced settings
- ☒ User Constants
- ☒ Config parameters
- ☒ Include parameters

Configure the below parameters :

Search (Ctrl+F)

- API
 - FreeRTOS API CMSIS v1
- Versions
 - FreeRTOS version 10.3.1

CMSIS-RTOS version

CMSIS-RTOS version

Diagnostics:

Pinout view System view

Pinout view



STM32F407VGTx
LQFP100

IDE day04 - Device Configuration Tool - STM32CubeIDE

File Edit Navigate Search Project Run Window Help

uart.ioc main.c

Pinout & Configuration

Categories A->Z

- USART1
- ✓ USART2
- USART3
- USART6
- USB_OTG_FS
- USB_OTG_HS

Multimedia >

Security >

Computing >

Middleware >

- FATFS
- ✓ **FREERTOS**
- LIBJPEG
- LWIP
- MBEDTLS
- PDM2PCM
- USB_DEVICE
- USB_HOST

Clock Configuration

Software Packs Pinout

FREERTOS Mode and Configuration

Mode

Interface CMSIS_V1

Configuration

Reset Configuration

✓ Mutexes	✓ Events	✓ FreeRTOS Heap Usage
✓ User Constants	✓ Tasks and Queues	✓ Timers and Semaphores
✓ Config parameters	✓ Include parameters	✓ Advanced settings

Configure the below parameters :

Search (Ctrl+F)

✓ Newlib settings (see parameter description first)

USE_NEWLIB_REENTRANT	Enabled
-----------------------------	---------

✓ Project settings (see parameter description first)

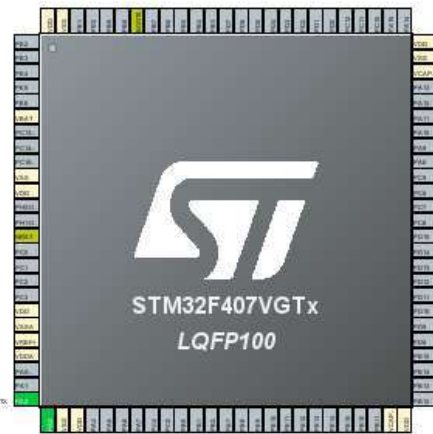
Use FW pack heap file	Enabled
-----------------------	---------

USE_NEWLIB_REENTRANT
USE_NEWLIB_REENTRANT
Parameter Description:

Project Manager

Tools

Pinout view System view



100% 13:27

Pinout & Configuration

Clock Configuration

Project Manager

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Software Packs

Pinout

FREERTOS Mode and Configuration

Configuration

Reset Configuration

☒ Timers and Semaphores

☒ Mutexes

☒ Events

☒ FreeRTOS Heap Usage

☒ Advanced settings

☒ User Constants

☒ Tasks and Queues

☒ Config parameters

☒ Include parameters

Tasks

Task Name	Priority	Stack Siz...	Entry Fun...	Code Gen...	Parameter	Allocation	Buffer Name	Control Bl...
defaultTask	osPriority...	128	StartDefa...	Default	NULL	Dynamic	NULL	NULL

Edit Task

Task Name

defaultTask

Priority

osPriorityIdle

Stack Size (Words)

128

Entry Function

StartDefaultTask

Code Generation Option

Default

Parameter

NULL

Allocation

Dynamic

Buffer Name

NULL

Control Block Name

NULL

OK

Cancel

Add

Delete

Add

Delete

Pinout view

System view

STM32F407VGT
LQFP100

Pinout & Configuration

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Software Packs

Pinout

FREERTOS Mode and Configuration

Configuration

Reset Configuration

☒ Timers and Semaphores	☒ Mutexes	☒ Events	☒ FreeRTOS Heap Usage
☒ Advanced settings	☒ User Constants	☒ Tasks and Queues	
☒ Config parameters	☒ Include parameters		

Configure the below parameters :

Search (Ctrl+F)

Include definitions

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Disabled
xSemaphoreGetMutexHolder	Disabled
osTaskGetTaskName	Disabled

vTaskDelayUntil

vTaskDelayUntil

Parameter Description:

Pinout view

System view

STM32F407VGTx
LQFP100

- IDE leds
 - Includes
 - Core
 - Inc
 - Src
 - Startup
 - Drivers
 - CMSIS
 - Device
 - Include
 - STM32F4xx_H
 - leds.ioc
 - STM32F407VGTx
 - STM32F407VGTx

- Categories A-Z
- System Core
 - Analog
 - Timers
 - Connectivity
 - Multimedia
 - Security
 - Computing
 - Middleware
 - FATFS
 - ☒ FREERTOS
 - LIBJPEG
 - LWIP
 - MBEDTLS
 - PDM2PCM
 - USB_DEVICE
 - USB_HOST

Software Packs

Pinout

Pinout view System view

FREERTOS Mode and Configuration

Mode

Interface CMSIS_V1

IDE Question

Do you want generate Code?

☒ Remember my decision

Yes No

Tasks and Queues Timers and Semaphores

Advanced settings User Constants

Config parameters Include parameters

Configure the below parameters :

Search (Ctrl+F)

API

FreeRTOS API CMSIS v1

Versions

FreeRTOS version 10.3.1

CMSIS-RTOS version 1.02

MPU/FPU

ENABLE_MPU Disabled

ENABLE_FPU Disabled

